

Lucas Alexander

Harvard, MA | lucashenryalexander@gmail.com | 978 - 798 -0979

Github: Lucasxyz5, LinkedIn: www.linkedin.com/in/lucas-alexander-40a10834 Website: <https://lucasxyz5.github.io/Lucasxyz5-Website/>

Education:

Tufts University

Masters of Science | Computer Science

Somerville, MA | Expected Spring 2028

Brandeis University

Bachelor of Science | Computer Science

Waltham, MA | Fall 2022 - Spring 2026

The Bromfield School

Harvard, MA | 08/2016 - 06/2022

Skills:

Technical Skills:

Languages: Java, Python, C++, SQL, Javascript, HTML

Programs/Libraries: Autodesk AutoCAD 24.1, Autodesk Fusion 360, Adobe Photoshop CC, Adobe InDesign, Microsoft Office Suite (Word, Excel, Powerpoint), Google Workspace, WxWidgets with C++, Chrono with C++, SFML with C++, Concurrency in Java

General: Knowledge of electronics and robotics, experience with calculus and linear algebra

Other Skills:

Languages: Spanish

English: Creative Writing, Copy Editing

Independent Projects:

Modding Experience | September 2018, 2024 - Present

- Created mods for games “Mount and Blade: Warband” and “Vintage Story”
- Built using pre-existing tools, online tutorials, and JSON
- Uploaded to VSMoDB and Nexus

Computer Assembly | Summer 2022

- Assembled a computer from store-bought parts
- Troubleshooted technical problems when computer refused to start
- Installed Windows 11 from scratch using USB

Raspberry Pi Coding Projects | Winter 2024 - Summer 2025

- Assembled a raspberry pi from a kit
- Used it to interact with various other electronics using code, such as a relay or a stepper motor
- Assembled code from scratch to communicate with relay, subsequently published code to Github

Personal Website Creation | Summer 2025

- Created a personal website using HTML and Javascript
- Learned about how to post it to the internet using tools like NGINX
- Designed the UI by hand using CSS, HTML and Javascript

Pong | Fall 2026

- Created the classic game “Pong” using C++ and SFML
- Learned about the basics of graphics rendering on a 2d plane in the process
- Gained experience in quickly adapting to an unknown library

Academic Projects:

Website Creation, COSI 103A - Fundamentals of Software Engineering

Spring Semester 2024

- Made a website to display recipes over the course of an entire semester using Javascript and HTML
- Rotated positions in team: Served as team leader, Unit tester, and coder at various times
- Used Git, Docker and Microsoft Azure to aid in development

Data Analysis, COSI 104A - Introduction to Machine Learning

Fall Semester 2024

- Used python to construct machine learning algorithms to analyse a variety of datasets
- Found trends in the data and used those to predict outcomes
- Used techniques like K-nearest neighbor, logistic regression, and random forest

Concurrent Shell Program, COSI 131A - Fundamentals of Computer Systems

Fall Semester 2025

- Used java to create a shell-like program that can process basic regular expressions, including chaining commands together with pipes or output symbols

- Made commands run concurrently, making commands much faster and allowing larger commands to run in the background.
- Included meta-commands to view and terminate commands even before they finished

Additional Experience:

Arts Editor for Brandeis Hoot | *Brandeis*

Waltham, MA | 2024 - 2025

- Wrote articles for the arts section of The Brandeis Hoot
- Handled newspaper layout using Adobe InDesign for the arts section
- Communicated with new writers and reviewed the articles they sent in for the arts section

Member of Robotics Club | *Brandeis*

Waltham, MA | 2024 - present

- Used Autodesk Fusion 360 to design and model various objects, trained for use of 3D printer
- Assembled remote controlled car using 3d printed parts
- Helped another member program a robot to play poker

Electronics assembly with Raspberry Pi | *Home*

Harvard, MA | Summer 2025

- Created electronic circuits for use with a Raspberry Pi, utilizing infrared sensors, actuating motors, and connecting to serial ports
- Used multimeters and other similar tools to diagnose problems with circuits
- Studied both electronics and the math behind circuits both independently and academically

Work History:

Instructor | iDtech

Waltham, MA | Summer 2026

- Taught ages 13 - 17 how to create, use and code robots in C++ for mock battles
- Solved problems as they appeared, whether they be mechanical, digital, or social
- Created and coordinated activities and challenges for the kids

Other Activities:

Member of Chess Club | *The Bromfield School*

Harvard, MA | 2020

Member of Speech and Debate Club | *The Bromfield School*

Harvard, MA | 2019

Member of Long Distance Running Club | *The Bromfield School*

Harvard, MA | 2018

Volunteer at Habitat for Humanity | *Leominster*

Harvard, MA | 2022